Teacher Wu Kuan | See It at a Glance #8 — $\sin^2\theta + \cos^2\theta = 1$

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1. The trap question

Why is $\sin^2\theta + \cos^2\theta$ always 1?

2. The method

- Unit circle radius 1; point at angle $\theta = (\cos\theta, \sin\theta)$.
- Legs $\cos\theta$ and $\sin\theta$, hypotenuse 1 → $\cos^2\theta + \sin^2\theta = 1$.
- Use it to find one ratio from the other.

3. Parent check method

Ask your child to point to “cos” and “sin” on the drawn triangle.

Practice

1. $\sin^2 37° + \cos^2 37° = ?$
2. θ acute, $\cos\theta = 4/5$. Find $\sin\theta$.
3. θ acute, $\sin\theta = 12/13$. Find $\cos\theta$.
4. θ acute, $\sin\theta = 0.8$. Find $\cos\theta$.
5. Simplify $1 - \sin^2\theta$.

Answers

1. 1
2. 3/5
3. 5/13
4. 0.6
5. $\cos^2\theta$

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